



XXI Ecological Integration Symposium

Ecology in the Anthropocene

Virtual Symposium
Texas A&M University

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College Station,
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PROGRAM BOOK



TEXAS A&M UNIVERSITY
Ecology and Conservation
Biology



TEXAS A&M UNIVERSITY
Recreation, Park
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TEXAS A&M UNIVERSITY
Animal Science



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Effect of the heritable bacterium *Spiroplasma* on the ability of its host *Drosophila* to counter parasitism by two wasp species

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There is an interaction between *Drosophila* and *Spiroplasma*. *Drosophila* maternally inherit bacteria from members of this *Spiroplasma* genus, and the prevalence of *Spiroplasma* varies by species and population (reviewed in Mateos et al., 2016). Three distantly related species of *Drosophila* (*D. neotestacea*, *D. hydei* and *D. melanogaster*) have shown to have an important relationship with *Spiroplasma* as the *Spiroplasma* helps defend against natural enemies. (reviewed in Mateos et al., 2016). Three of the *Spiroplasma* strains Shy1, MSRO, and Sneo (which belong to the same lineage: poulsonii clade) are found in *D. hydei*, *D. melanogaster*, and *D. neotestacea*, respectively, and are known to protect against parasitic wasps (and the parasitic nematode in *D. neotestacea*) (reviewed in Mateos et al., 2016). This inhibition caused by the *Spiroplasma* suggests that *Spiroplasma* is able to hinder the development of larval parasitoids. This defense is considered to happen through a combination of RIP toxins that are secreted by the *Spiroplasma* and also a competition for lipids between the *Spiroplasma* and the wasp (Jones and Hurst, 2020). However, past studies have revealed that resistance to *Spiroplasma* has likely evolved at least once in *Leptopilina* sp. and *Ganaspis* sp. (reviewed in Mateos et al., 2016). With this, the proposed study will examine the effects of *Spiroplasma* on the outcome of parasitism by two previously uncharacterized wasp species: *Ganaspis* sp. (M. Mateos, pers. comm.) and *Leptopilina* aff. *L. clavipes*. (M. Mateos, pers. comm.).

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Effect of wasp parasitism on the bacterial densities in the *Spiroplasma-Drosophila melanogaster* symbiosis

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Most insect species are associated with microorganisms residing within their tissues, known as endosymbionts. Bacteria of the genus *Spiroplasma* occur naturally in the hemolymph of many species of *Drosophila*, including *Drosophila melanogaster*, where they are vertically transmitted from mother to offspring. The presence of these endosymbionts can have diverse effects on their hosts, either at adult or larval stages. Strains of *Spiroplasma* have been shown to protect *Drosophila* larvae against parasitic wasps, although the effect varies with wasp species and strain. The mechanisms by which certain wasp species resist or tolerate the effect of *Spiroplasma* are not completely understood. Bacterial endosymbiont density has been shown to influence certain phenotypes. In this study, *Spiroplasma*-infected fly larvae were exposed to different parasitoid wasps in order to determine whether wasp resistance involves a change in the growth of *Spiroplasma* as estimated by quantitative PCR. The results showed consistently low *Spiroplasma* densities over time in flies exposed to *Leptopilina guineaensis*, one of the resistant wasps. However, when exposed to *Ganaspis hookeri*, the other resistant wasp, the flies had *Spiroplasma* densities increasing over time, similarly to flies unexposed to wasps, and flies exposed to *Leptopilina heterotoma*, a *Spiroplasma*-susceptible wasp. These results suggest that the mechanisms by which *L. guineaensis* and *G. hookeri* avoid *Spiroplasma*-induced damage may differ.



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